Power

VSYS comes from the 12 V input: buck to +5V, then D4 (Power sheet). D2 keeps that 5 V off the USB port; USB alone runs it through D2 (no +5V on ribbons).

CC1/CC2 5.1k pull-downs: a USB-C host sees a USB 2.0 device and turns on 5 V.

Radio (RM2)

RM2 gSPI: GPIO43 = WL_ON + BT_ON, GPIO44 = CS, GPIO45 = DATA (DI direct, DO through R14, IRQ through R15), GPIO46 = CLK. Same circuit as the Pico 2 W, moved off GPIO23/24/25/29; set CYW43_DEFAULT_PIN_* in the board header. Hand-solder U5 (not stocked at LCSC). Keep copper out from under the antenna end.

Primary Flash

If using RP2354 (internal flash), remove R10. R9 can be fit to use U3 as secondary memory.

BOOTSEL Button

PSRAM

8 MB PSRAM (U4) on the second QSPI chip select (XIP_SS_N[1]), moved from GPIO0 to GPIO47. R13 pull-up keeps it deselected at power-up (GPIOs default to pull-downs). Fit R11 = 0R.

MH1 NPTH FD2.75mm SM3.15mm KO6mm
MH2 NPTH FD2.75mm SM3.15mm KO6mm
MH3 NPTH FD2.75mm SM3.15mm KO6mm
MH4 NPTH FD2.75mm SM3.15mm KO6mm

12MHz Crystal

RUN Button

Make sure R7 and R8 are close to RP2350

12 V in, 5 V buck

Power
File: Power.kicad_sch

RTC and headers

RTC: the DS3231 circuit from universal-strip-controller (U9), on I2C0: SDA = GPIO32, SCL = GPIO33, INT/SQW = GPIO34 (open drain, pulled up). 32KHZ and RST unused. CR1220 in a Keystone 3001 holder.

GPIO0-31 to the four output shrouds

Outputs
File: Outputs.kicad_sch

J2: spare GPIO35-42. UART1 TX/RX = GPIO36/37. GPIO40-42 = ADC0-2. J3: SWD in a Pico's debug pin order. GPIO0-31 go to the four output shrouds (Outputs sheet). GPIO43-47 are the RM2 and the PSRAM.

Note: S1 schematic revision indicates that this is the 1st revision of the schematic/BOM associated with the R4 PCB. R4-S1 therefore uniquely identifies the fully built PCB Assembly (PCBA). This is a change from the previous revision numbering to align with the internal Raspberry Pi product revisioning scheme

SCHEMATIC/BOM REVISION: S1

BOM is autogenerated from schematic, therefore schematic rev. == BOM rev. Keep Titleblock rev. in sync with above. Reset to 1 when creating new R num.

PCB/PWB REVISION: R4 <- PCB/PWB rev this schematic is for

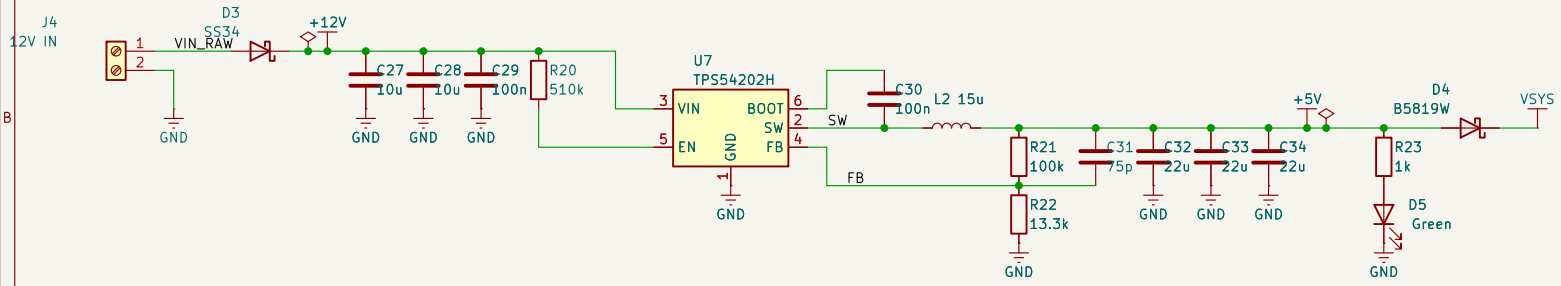
MANUFACTURER: XXX
MANUF. PN: XXX
CIRCUITS/PANEL: XXX
DESCRIPTION: XXX

VARIANT: VARIANT1
REF: RPI-RP2350B-MINIMAL
DESCR: RP2350B QFN-80 Minimal Design Example

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TITLE	LED processor board (RP2350B + RM2)
DATE	2026-10-14 REF: RP2350B-RM2 / RPI-RP2350B-MINIMAL REV: R4-S1
SIZE	A3 DRAWN: Original design: Mike Stimson, Raspberry Pi SHEET: 1/3

12 V in, 5 V buck



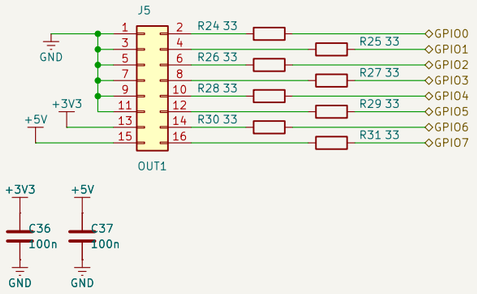
U7 values follow TI's TPS54202HEVM-716 5 V design (100k / 13.3k with 75 pF across the top resistor: 0.596 V x (1 + 100/13.3) = 5.08 V; 15 uH; 500 kHz). R20 enables U7 from the 12 V side. D3 blocks a reversed 12 V lead. The load is small: the 3.3 V regulator on VSYS (RP2350, RM2, flash, PSRAM) and the four output boards' shifters (VCCB on +5V, VCCA on +3V3), well under 0.5 A. +5V feeds the ribbons directly; VSYS gets it through D4 so USB can't back-feed the buck.

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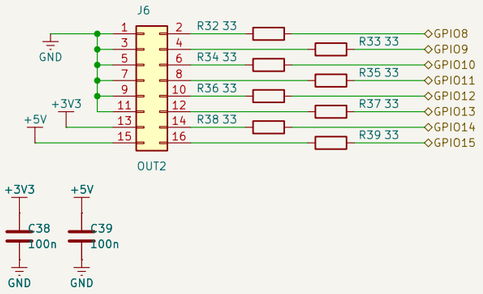
TITLE	LED processor board: Power				
DATE	2026-10-19	REF		REV	A
SIZE	A4	DRAWN		SHEET	2/3

Output shrouds: 16-pin IDC to the output boards. Pin 1 is the triangle on the shroud.
Odd pins 1-11 GND, 13 = +3V3 (shifter VCCA), 15 = +5V (shifter VCCB); data 1-8 on even pins 2-16
(same as led-output-board J2 and MegaTree docs/power.md). 33R series resistors at the RP2350 end.

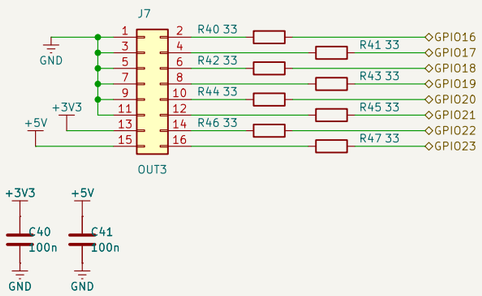
OUT1: GPIO0-7



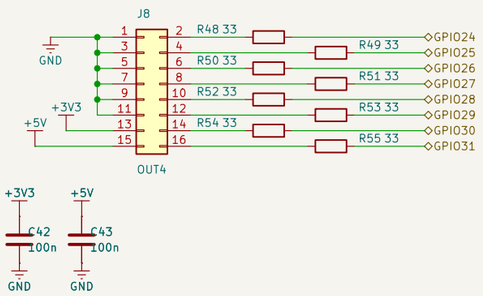
OUT2: GPIO8-15



OUT3: GPIO16-23



OUT4: GPIO24-31



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TITLE	LED processor board: Outputs				
DATE	2026-10-19	REF		REV	A
SIZE	A3	DRAWN		SHEET	3/3